Transformaciones 2D

(CAPÍTULO 2 – CV02B)

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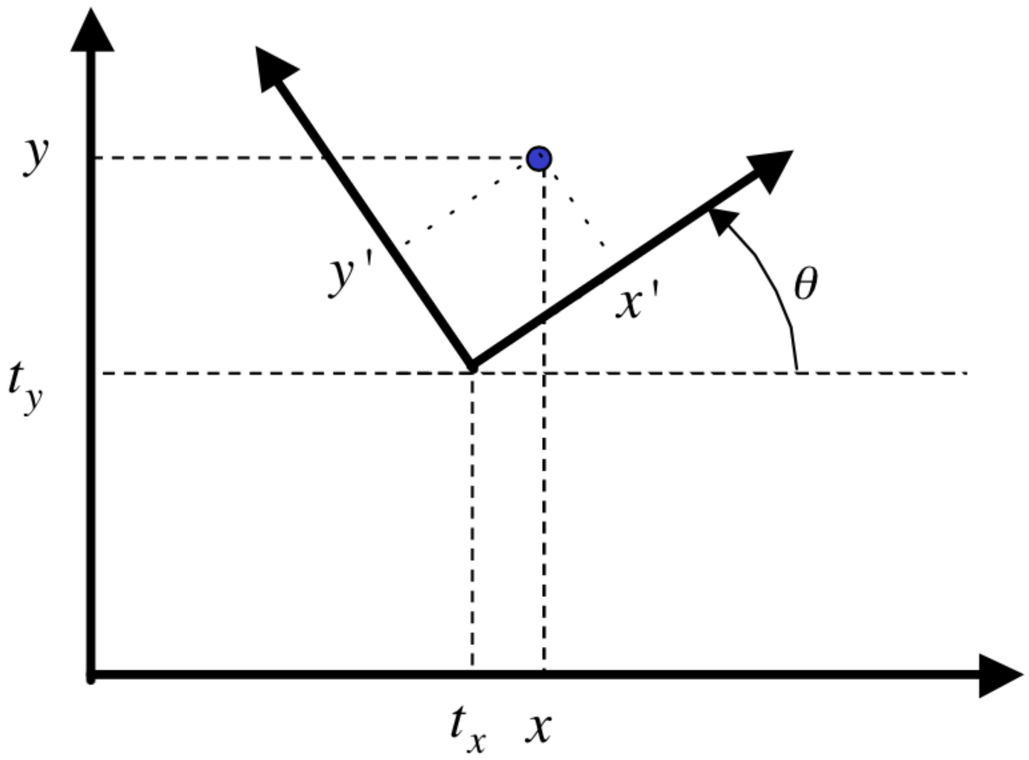
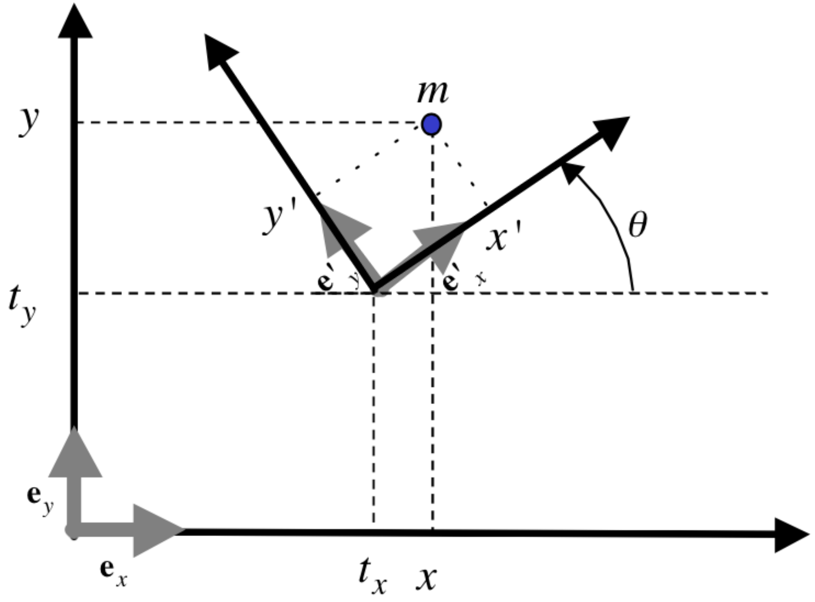
[Euclidean 2D Transformation]



Problem: Given (x',y') and (t_x, t_y, θ) find (x,y)

or: Given (x,y) and (t_x, t_y, θ) find (x',y')

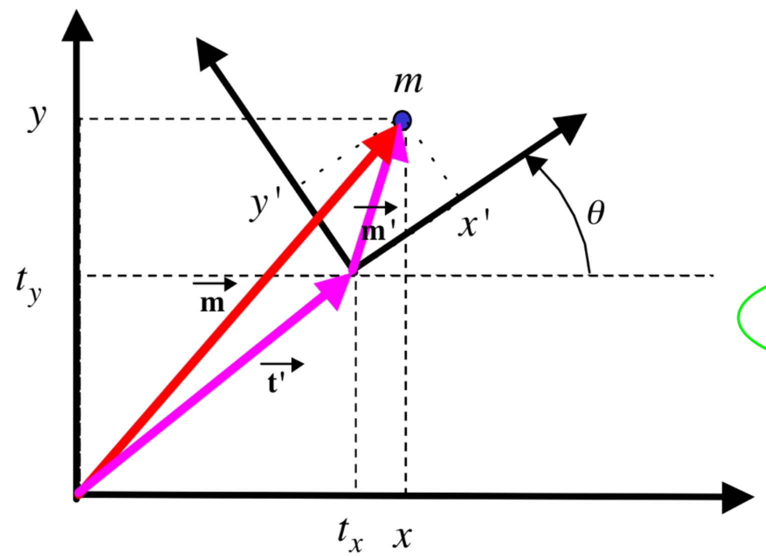
[Euclidean 2D Transformation]



$$\begin{aligned}\mathbf{e}'_x &= \cos(\theta)\mathbf{e}_x + \sin(\theta)\mathbf{e}_y \\ \mathbf{e}'_y &= -\sin(\theta)\mathbf{e}_x + \cos(\theta)\mathbf{e}_y\end{aligned}$$

$$\underbrace{x\mathbf{e}_x + y\mathbf{e}_y}_{\vec{\mathbf{m}}} = \underbrace{t_x\mathbf{e}_x + t_y\mathbf{e}_y}_{\vec{\mathbf{t}}'} + \underbrace{x'\mathbf{e}'_x + y'\mathbf{e}'_y}_{\vec{\mathbf{m}}'}.$$

[Euclidean 2D Transformation]



$$\begin{aligned} \mathbf{e}'_x &= \cos(\theta)\mathbf{e}_x + \sin(\theta)\mathbf{e}_y \\ \mathbf{e}'_y &= -\sin(\theta)\mathbf{e}_x + \cos(\theta)\mathbf{e}_y \end{aligned}$$

$$\underbrace{x\mathbf{e}_x + y\mathbf{e}_y}_{\vec{\mathbf{m}}} = \underbrace{t_x\mathbf{e}_x + t_y\mathbf{e}_y}_{\vec{\mathbf{t}}'} + \underbrace{x'\mathbf{e}'_x + y'\mathbf{e}'_y}_{\vec{\mathbf{m}}'}.$$

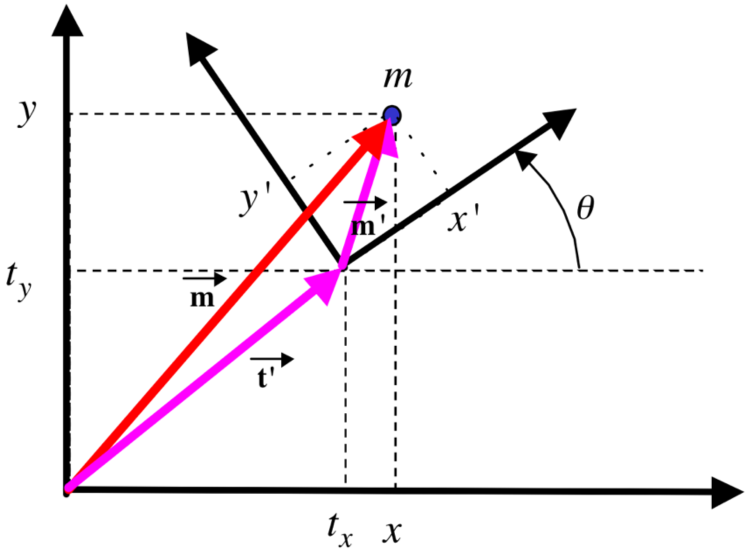
$$x\mathbf{e}_x + y\mathbf{e}_y = t_x\mathbf{e}_x + t_y\mathbf{e}_y + x'\cos(\theta)\mathbf{e}_x + x'\sin(\theta)\mathbf{e}_y - y'\sin(\theta)\mathbf{e}_x + y'\cos(\theta)\mathbf{e}_y.$$

$\mathbf{e}_x$ direction ➔ $$x = x'\cos(\theta) \quad -y'\sin(\theta) \quad +t_x$$

$\mathbf{e}_y$ direction ➔ $$y = x'\sin(\theta) \quad +y'\cos(\theta) \quad +t_y$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} = \mathbf{R}' \begin{bmatrix} x' \\ y' \end{bmatrix} + \mathbf{t}'$$

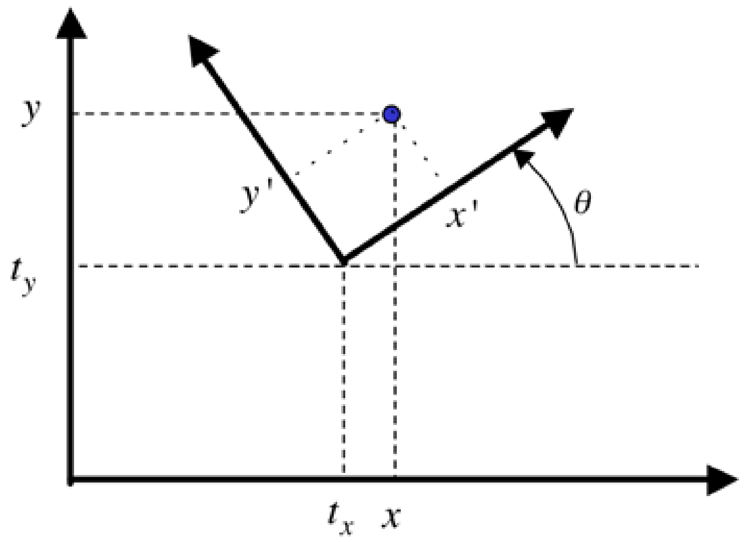
[Euclidean 2D Transformation]



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} = \mathbf{R}' \begin{bmatrix} x' \\ y' \end{bmatrix} + \mathbf{t}'$$

[Euclidean 2D Transformation]



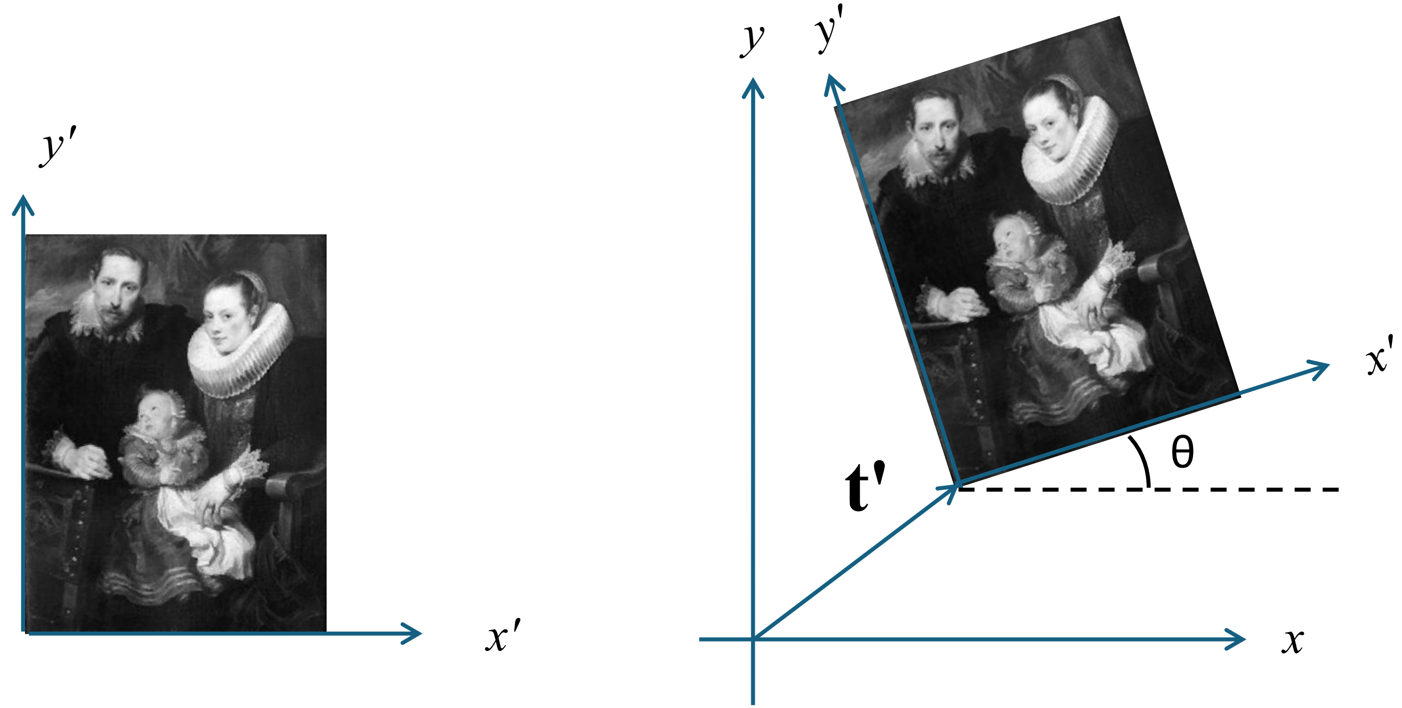
$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Problem: Given (x',y') and (t_x, t_y, θ) find (x,y)

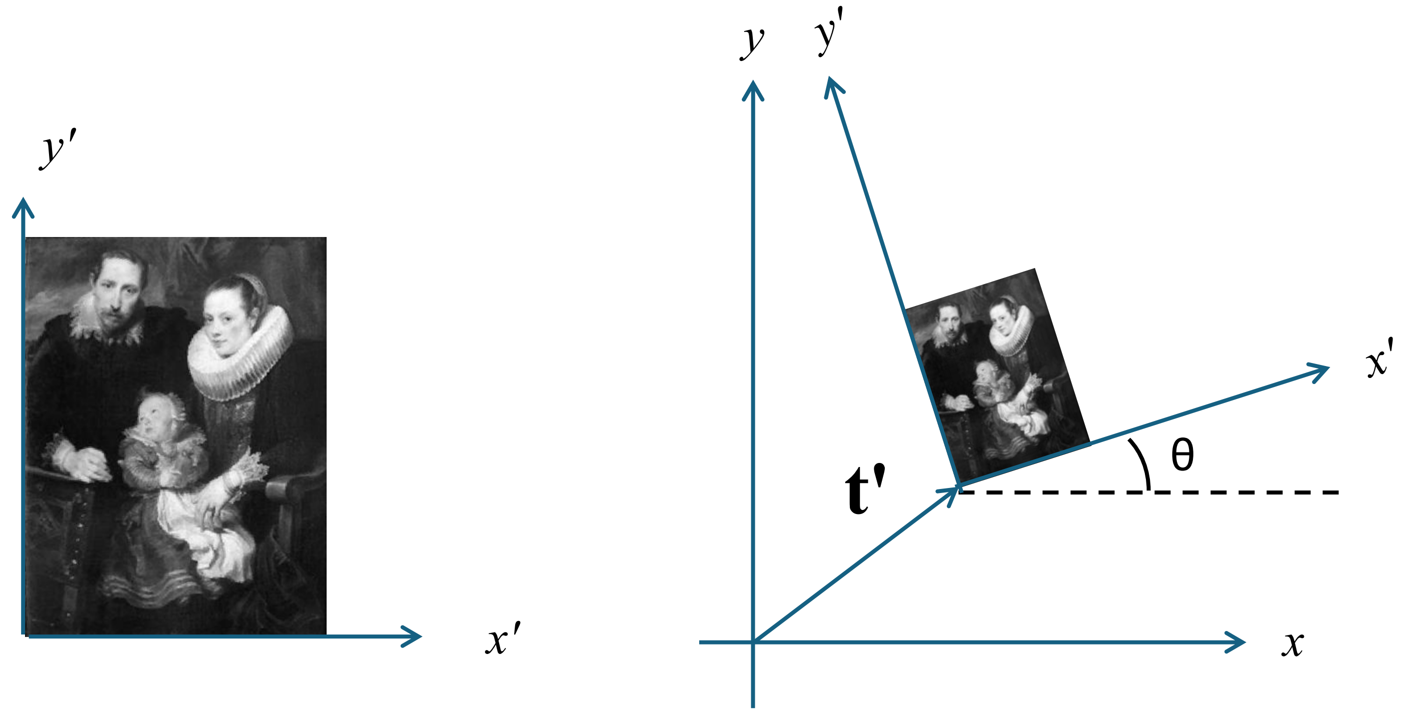
or: Given (x,y) and (t_x, t_y, θ) find (x',y')

[Euclidean 2D Transformation]



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

[Similarity 2D Transformation]



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s\mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

[2D Transformations – H matrix]

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

[2D Transformations – H matrix]

Euclidean

Similarity

Affine

General

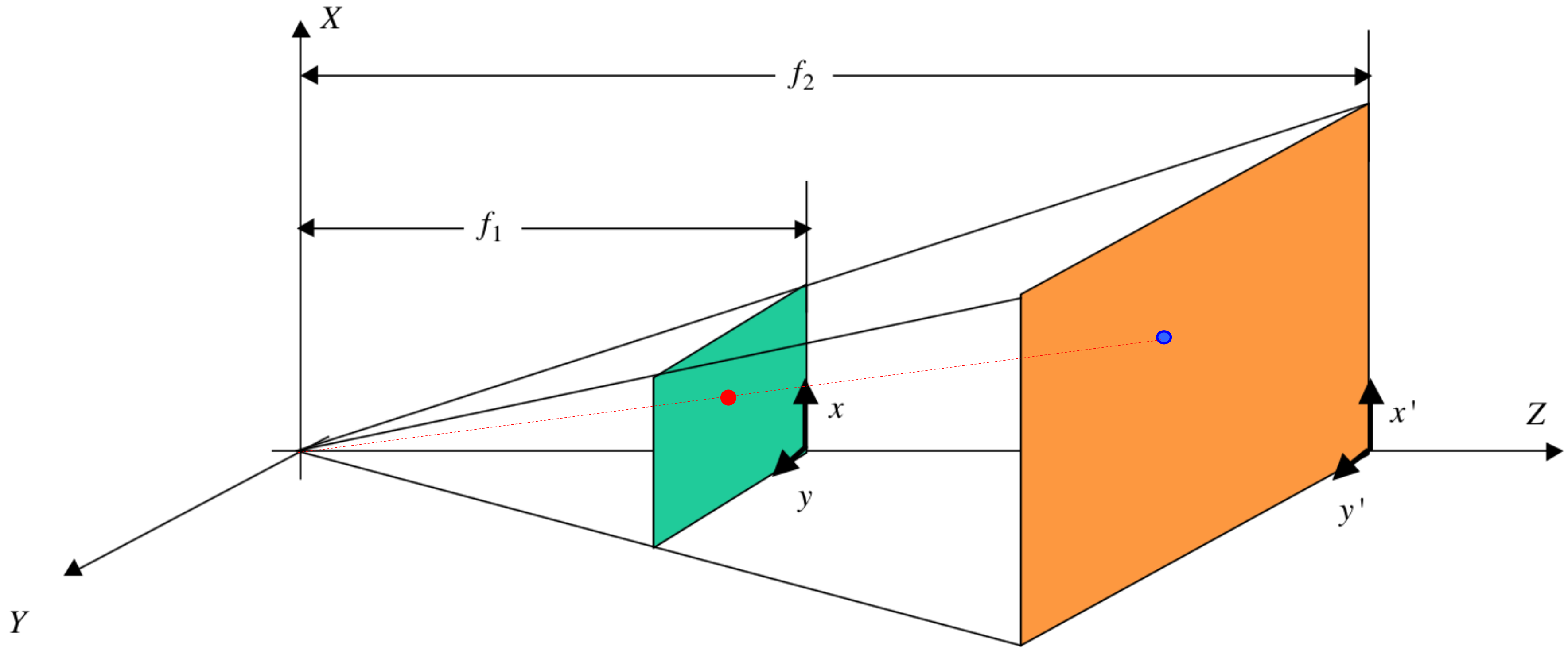
$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} s\cos(\theta) & -s\sin(\theta) & t_x \\ s\sin(\theta) & s\cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

[Homography]

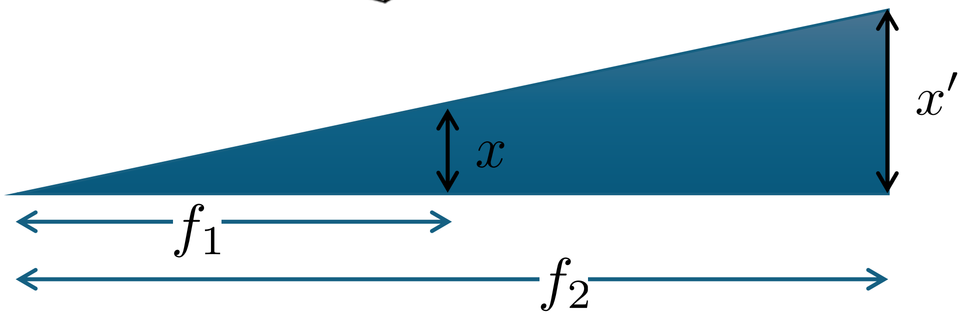
Transformation from two 2D spaces, where straight lines are transformed as straight lines.

[Homography: Example]

Problem: Given (x , y), find (x' , y')
(planes and are parallel and perpendicular to Z).



$$\frac{x'}{f_2} = \frac{x}{f_1} \qquad \frac{y'}{f_2} = \frac{y}{f_1}$$



[Homography: Example]

$$\frac{x'}{f_2} = \frac{x}{f_1} \qquad \frac{y'}{f_2} = \frac{y}{f_1}$$

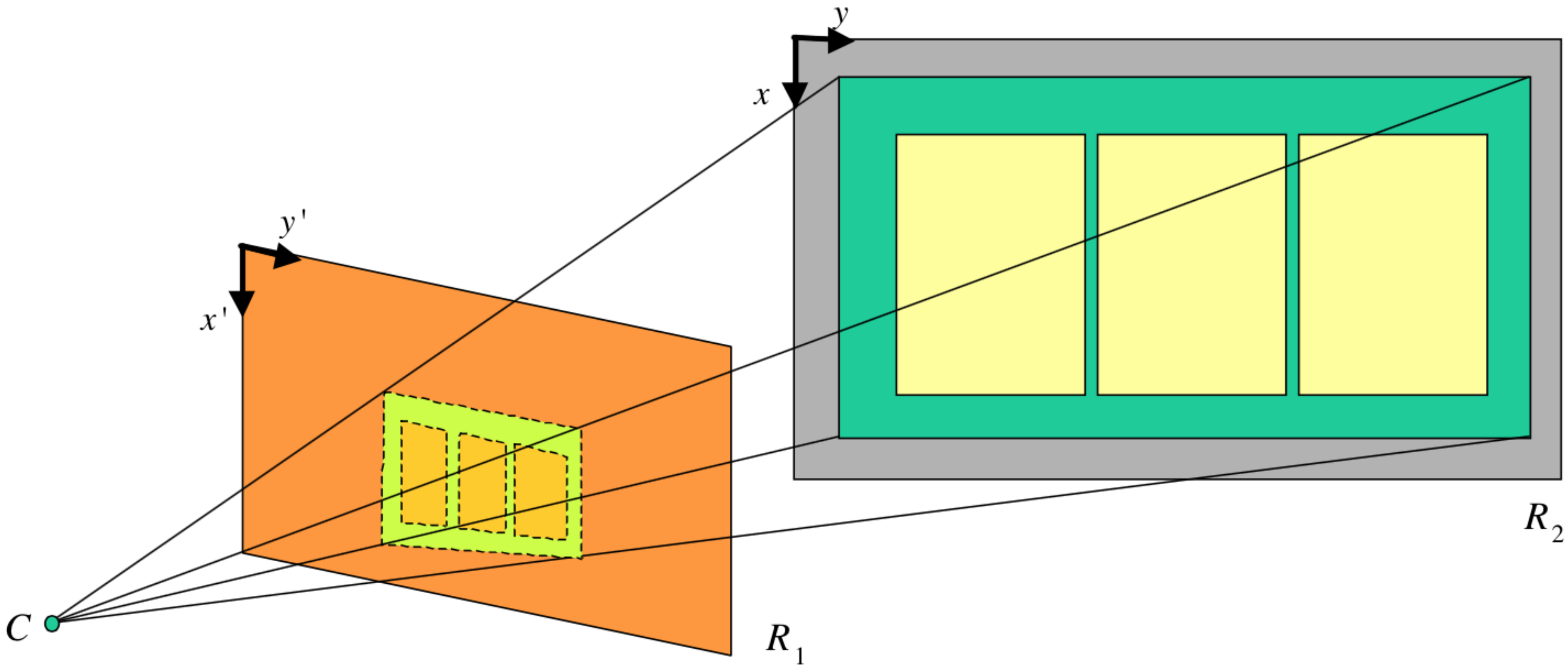
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} f_2/f_1 & 0 & 0 \\ 0 & f_2/f_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{m}' = \mathbf{H}\mathbf{m} \qquad \mathbf{H} = \begin{bmatrix} f_2/f_1 & 0 & 0 \\ 0 & f_2/f_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[Homography]

Transformation from two 2D spaces, where straight lines are transformed as straight lines.

[Homography: Planes R$_1$ and R$_2$ are not parallel]

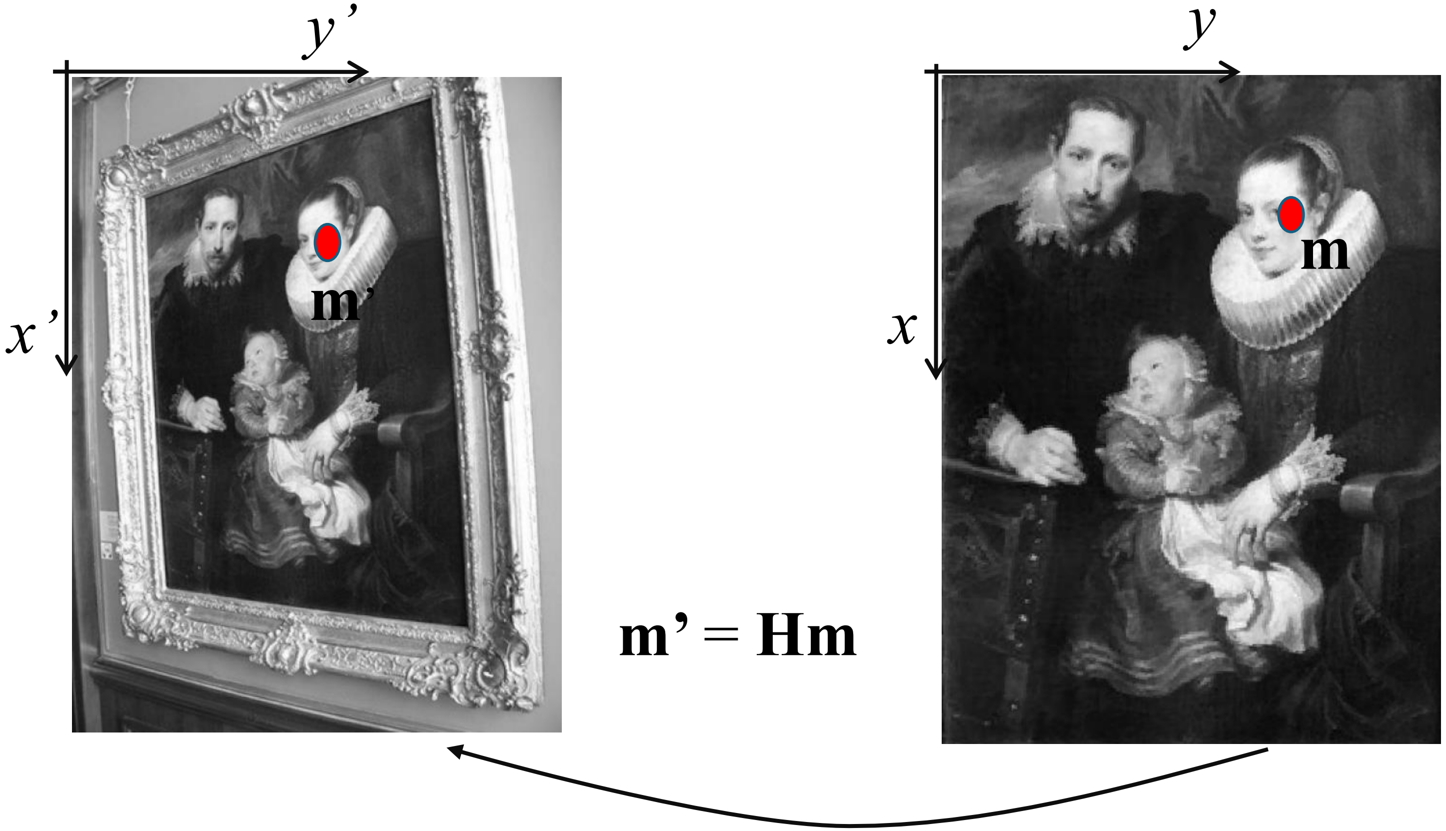


$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

[Homography]

Demo: pág 21 (D.Mery: Apuntes de Visión por Computador)

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



[Homography]

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\downarrow$$

$$x' = \frac{x_1'}{x_3'} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{x_2'}{x_3'} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

[Homography]

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

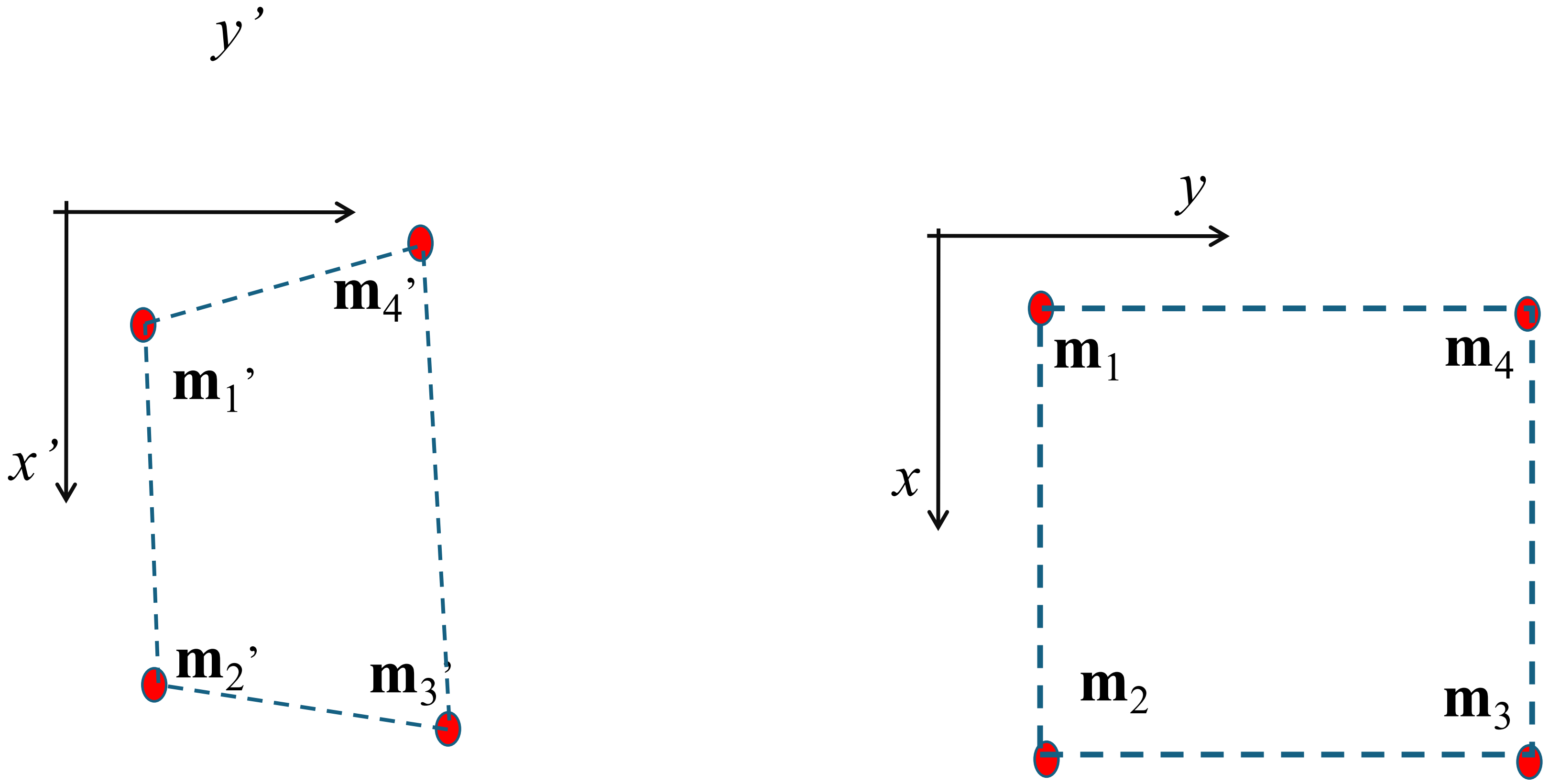
$$y' = \frac{x'_2}{x'_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

with $h_{33} = 1$

$$\underbrace{\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix}}_{\mathbf{h}} = \underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix}}_{\mathbf{b}}$$

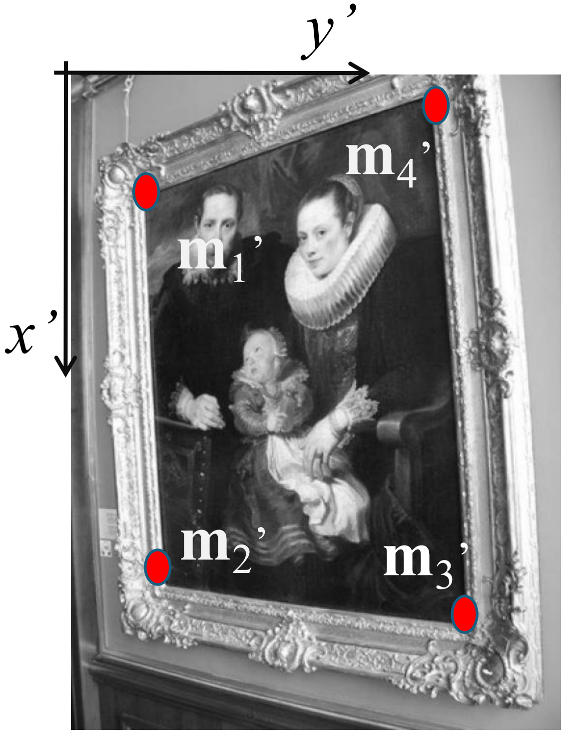
$$\mathbf{Ah} = \mathbf{b}$$

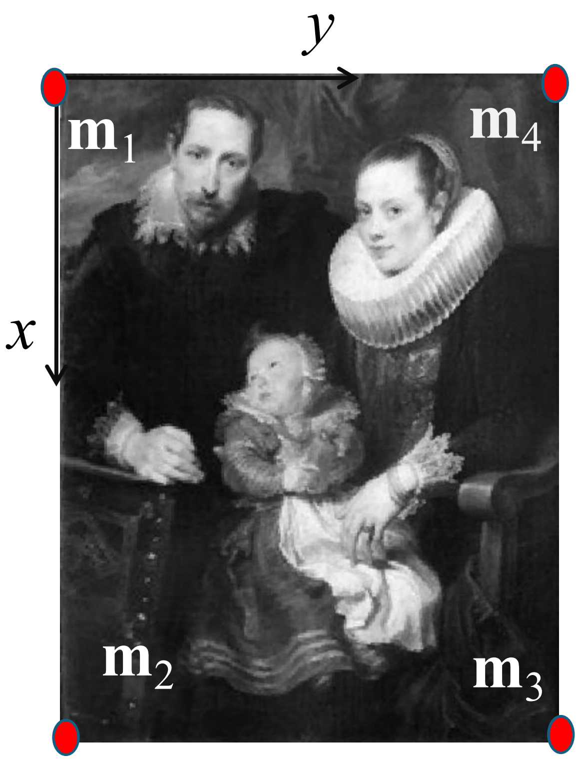
[Homography]



y'
x'
$\mathbf{m}_1$'
$\mathbf{m}_2$'
$\mathbf{m}_3$'
$\mathbf{m}_4$'
y
x
$\mathbf{m}_1$
$\mathbf{m}_2$
$\mathbf{m}_3$
$\mathbf{m}_4$

[Homography]





$$\underbrace{\begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_n \end{bmatrix}}_{\tilde{\mathbf{A}}} \mathbf{h} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_n \end{bmatrix}$$

$\mathbf{h} = \tilde{\mathbf{A}}^{-1}\mathbf{b}$ for $n = 4$

$\mathbf{h} = [\tilde{\mathbf{A}}^{\mathsf{T}}\tilde{\mathbf{A}}]^{-1}\tilde{\mathbf{A}}^{\mathsf{T}}\mathbf{b}$ for $n > 4$

Implementation

SOURCE

TARGET

I'

H ←

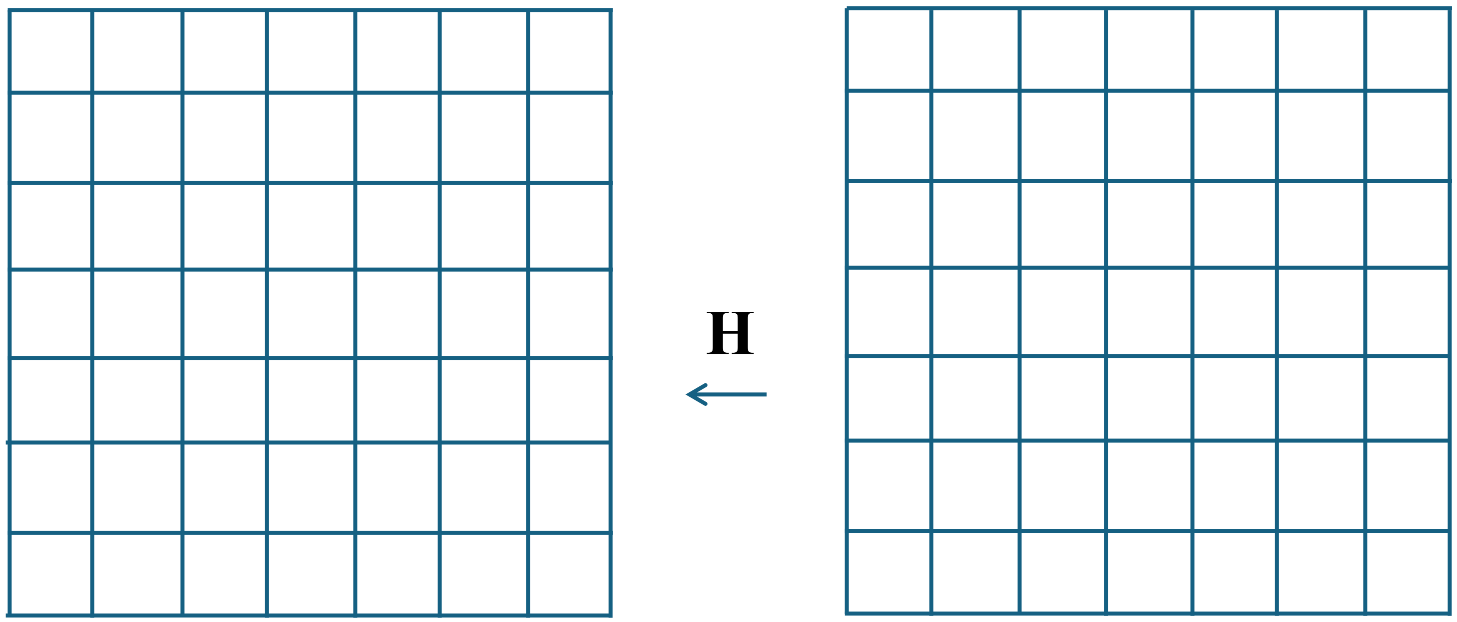
I

SOURCE

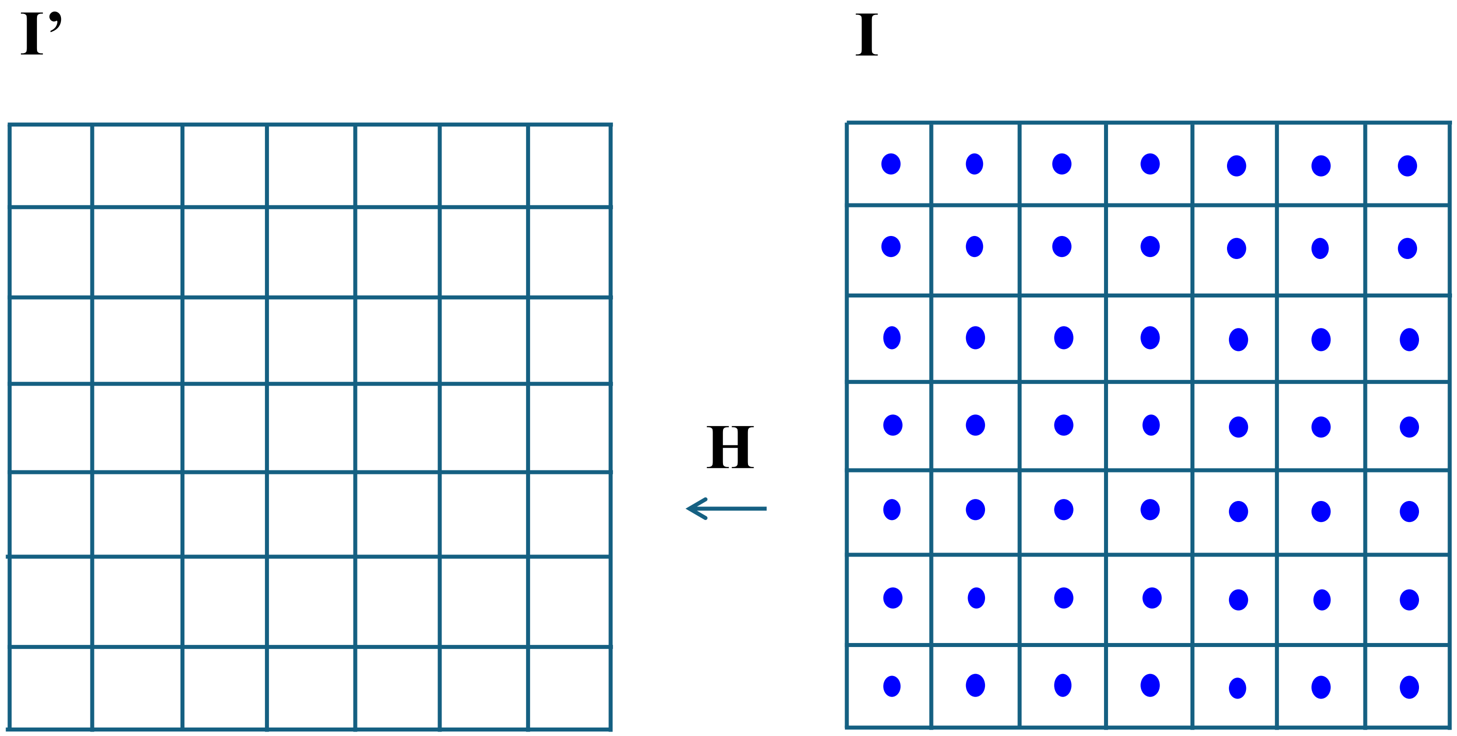
TARGET

I’

I



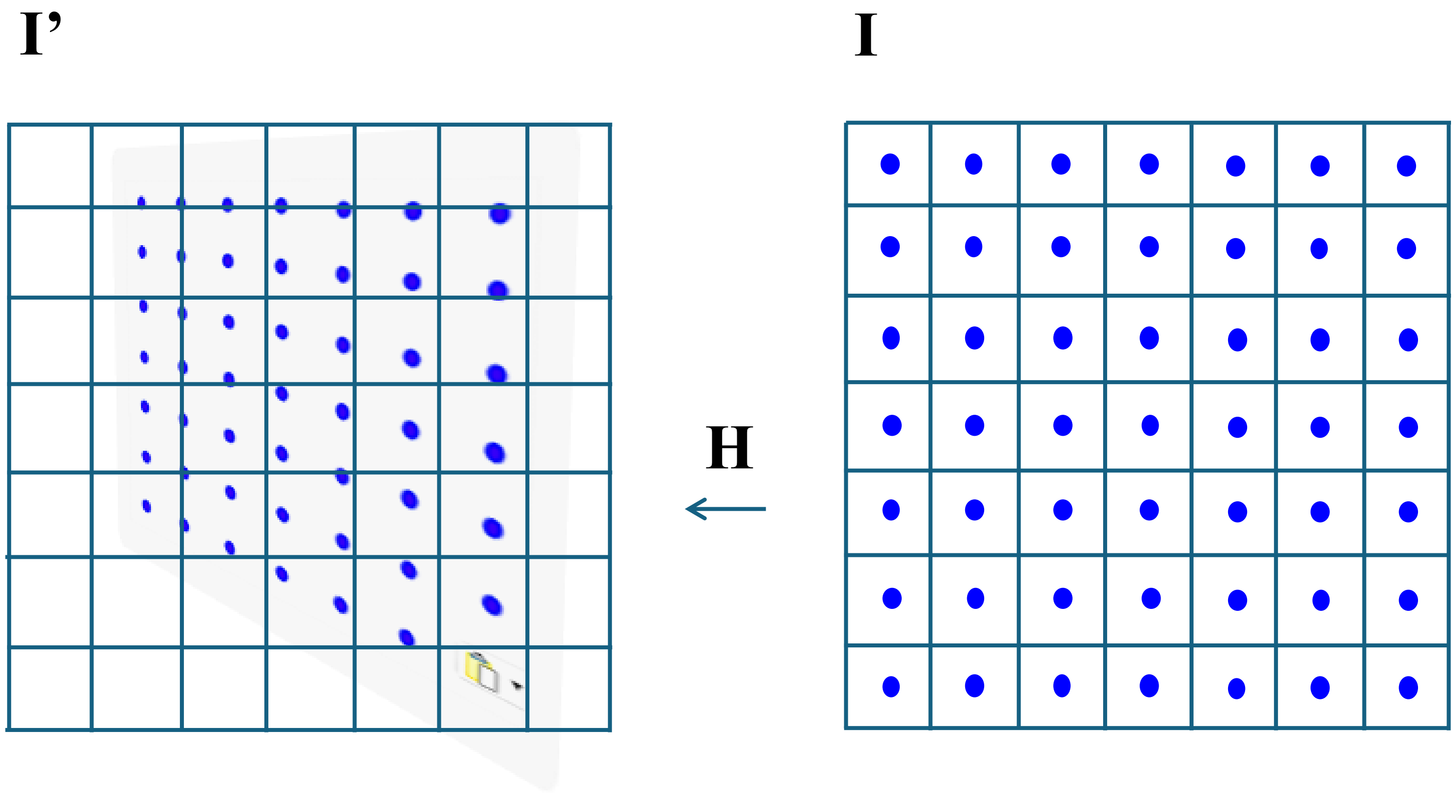
SOURCE TARGET



ALGORITHM: For each pixel in TARGET

SOURCE

TARGET

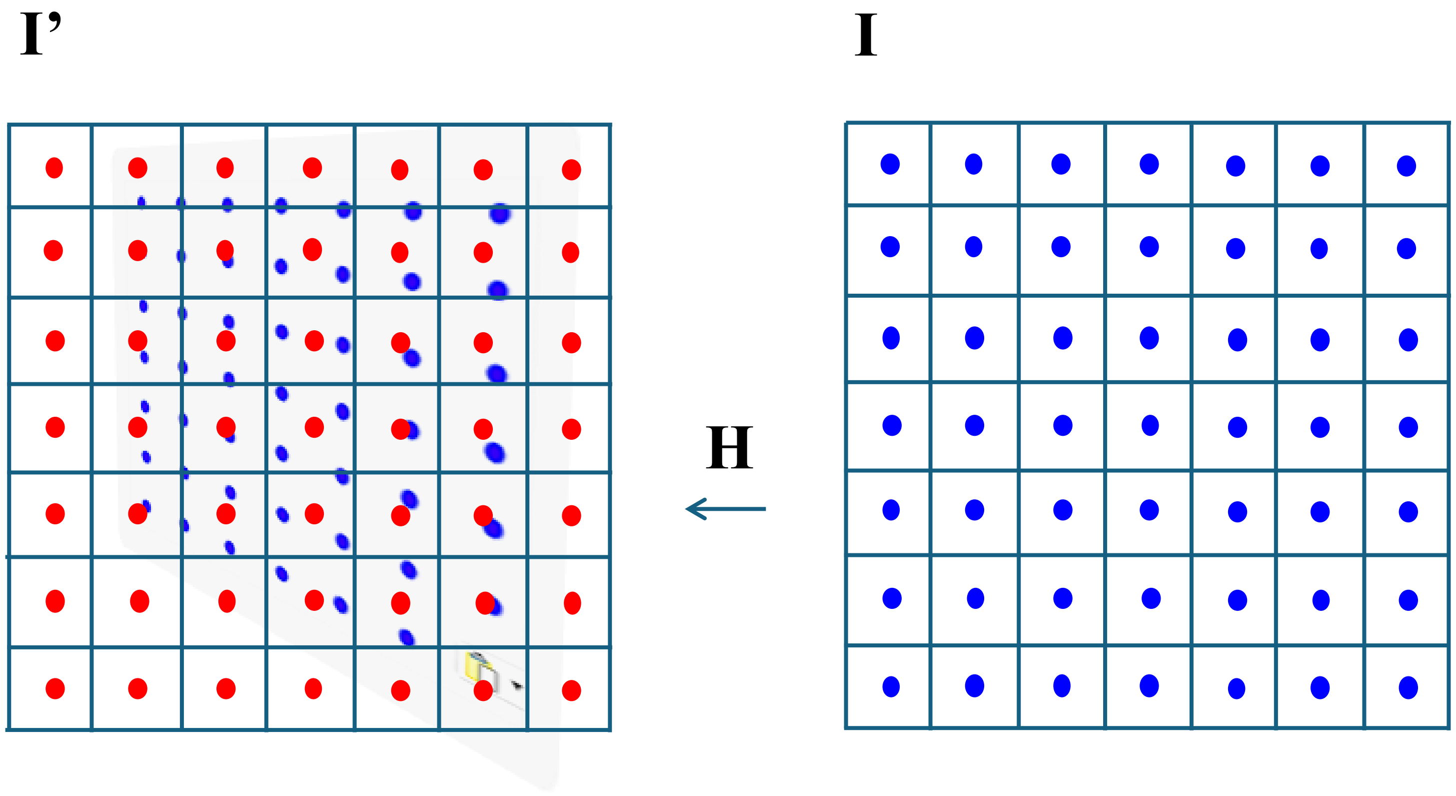


ALGORITHM: For each pixel in TARGET

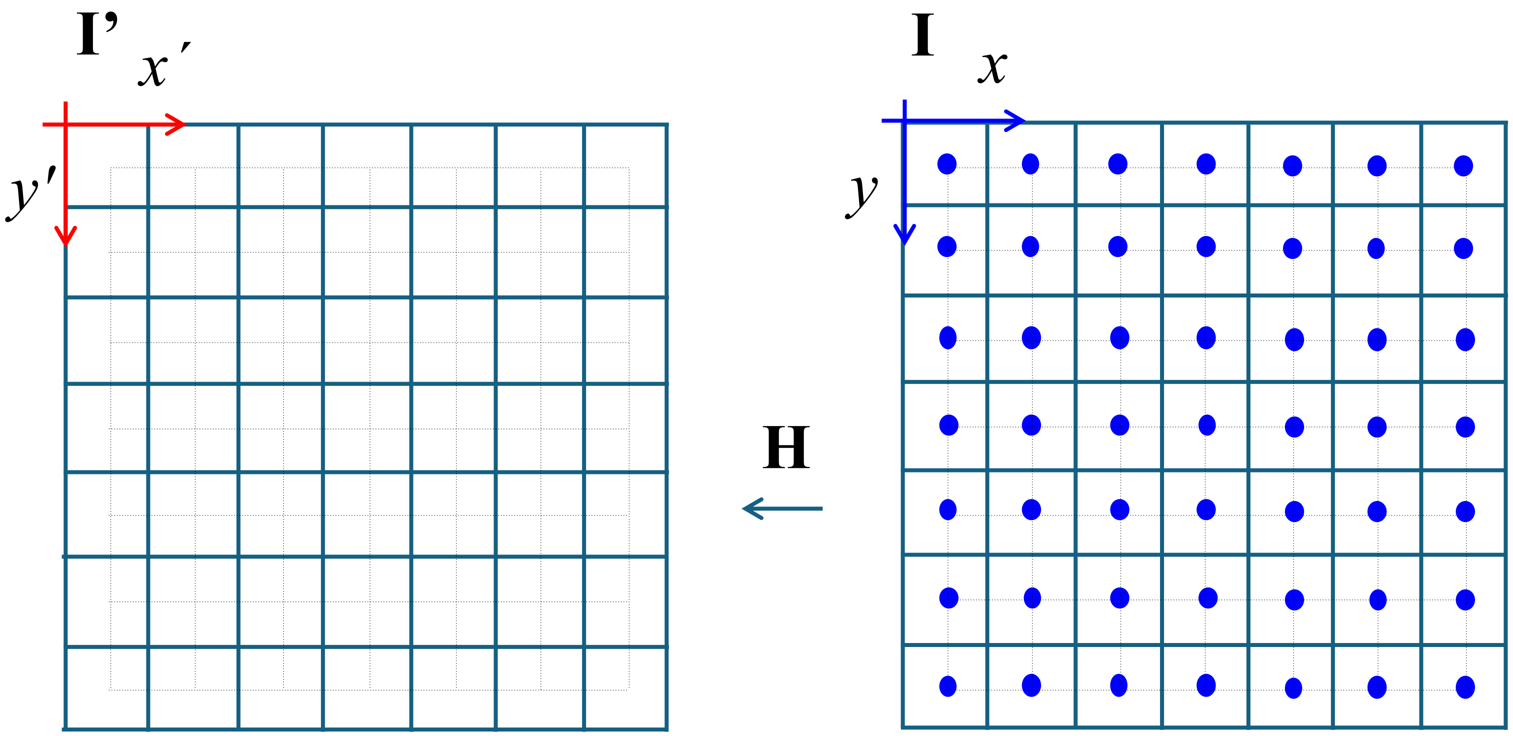
- compute the corresponding position in SOURCE

SOURCE

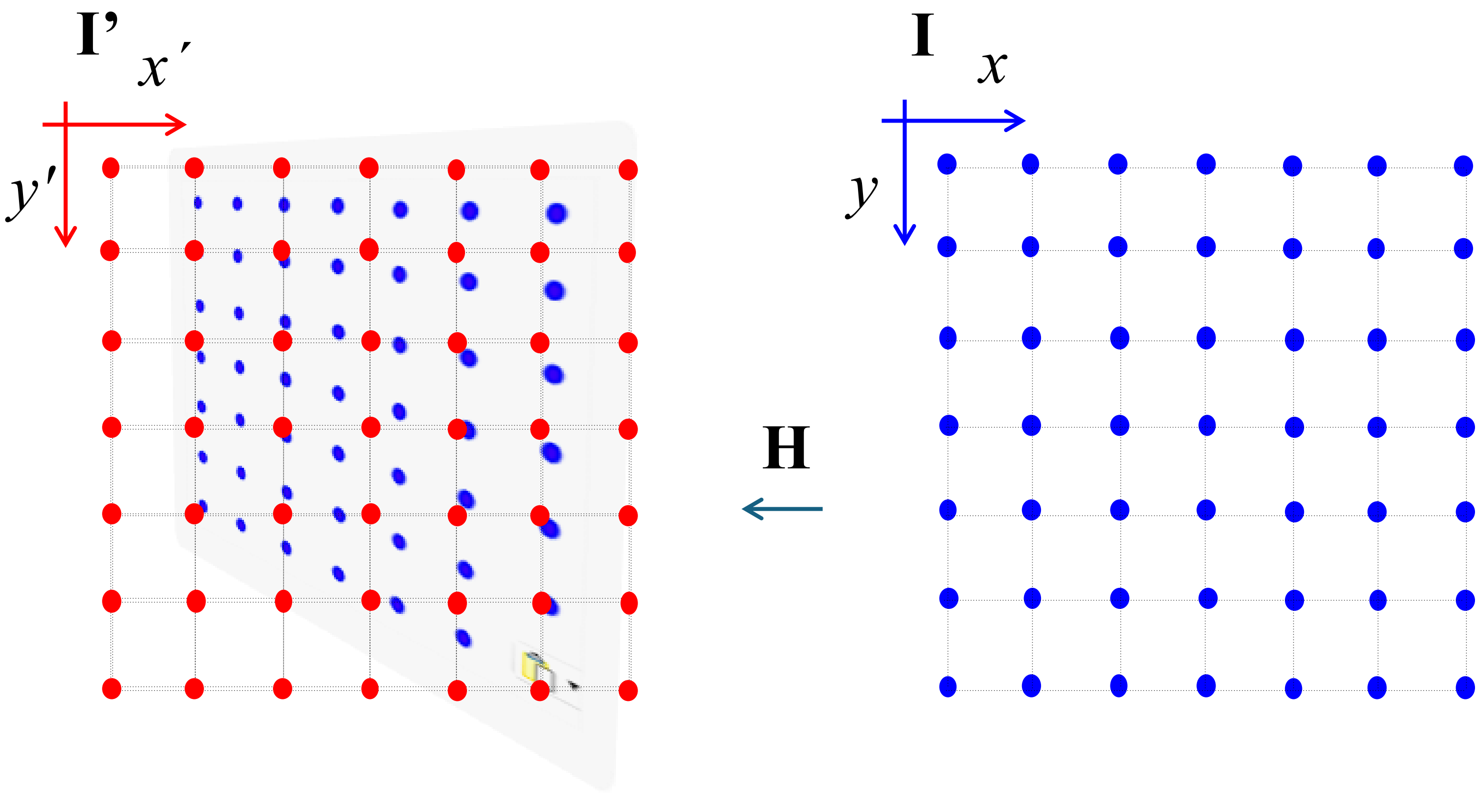
TARGET



ALGORITHM: For each pixel in TARGET
- compute the corresponding position in SOURCE
- interpolate the grayvalue using SOURCE pixels.

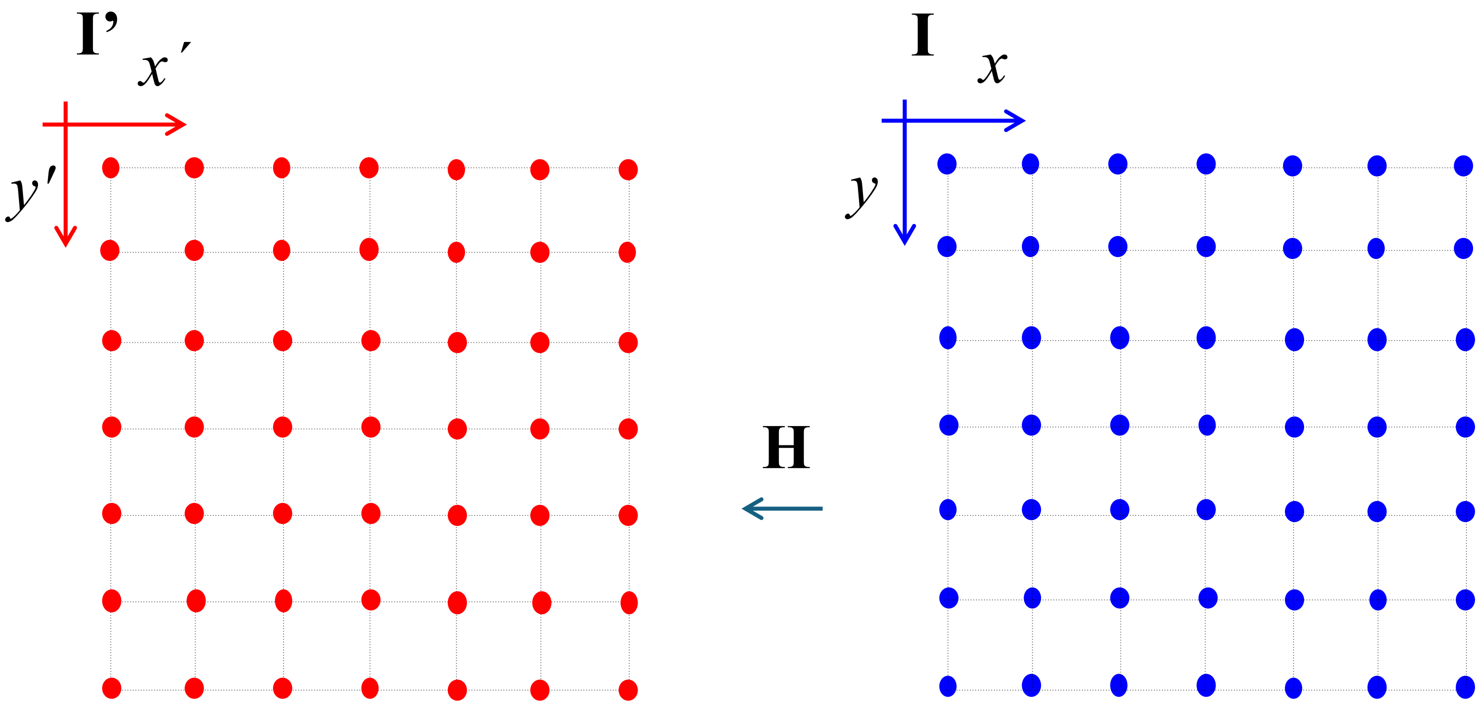


I'
x'
y'
H
I
x
y



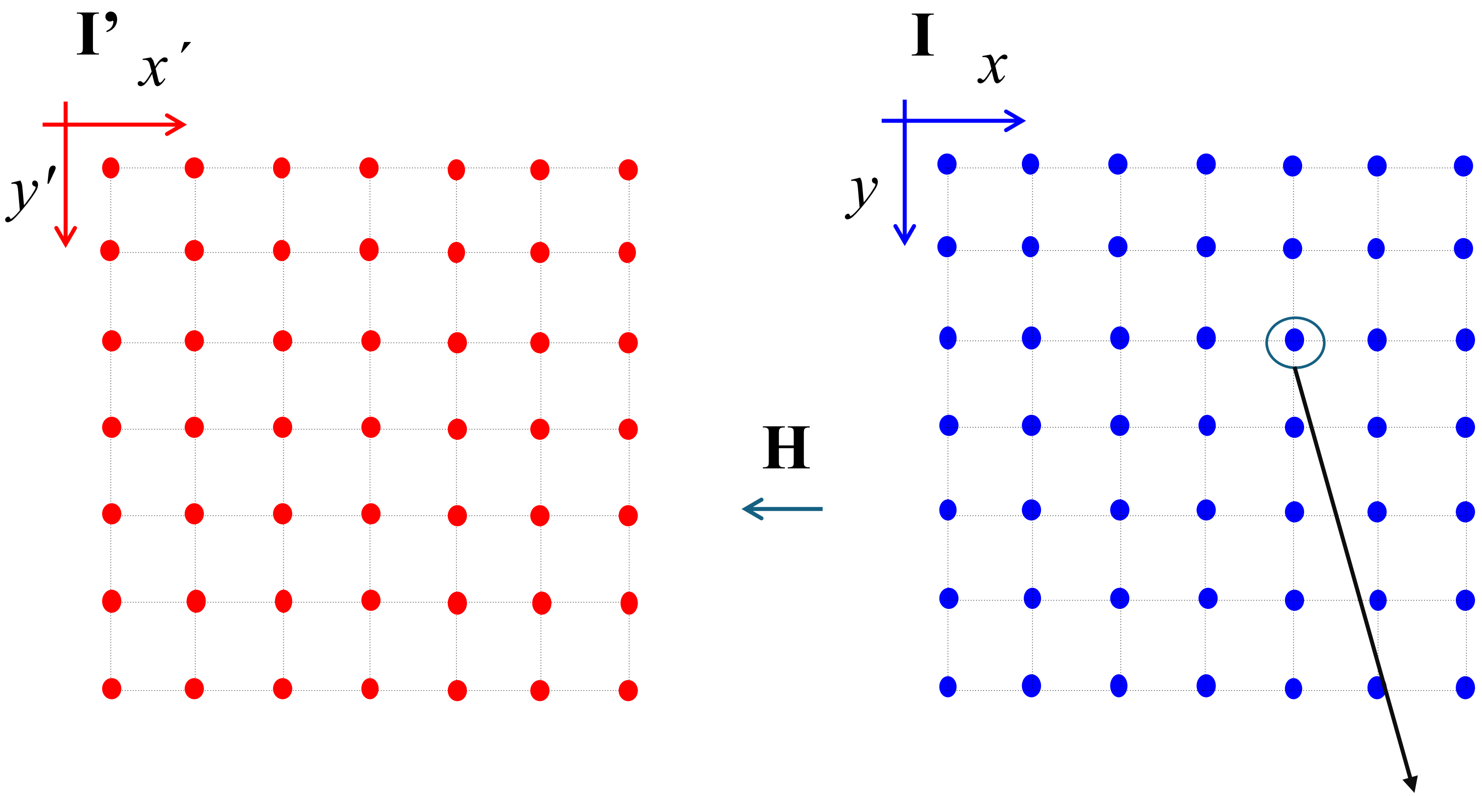
Coordinate Transformation:

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



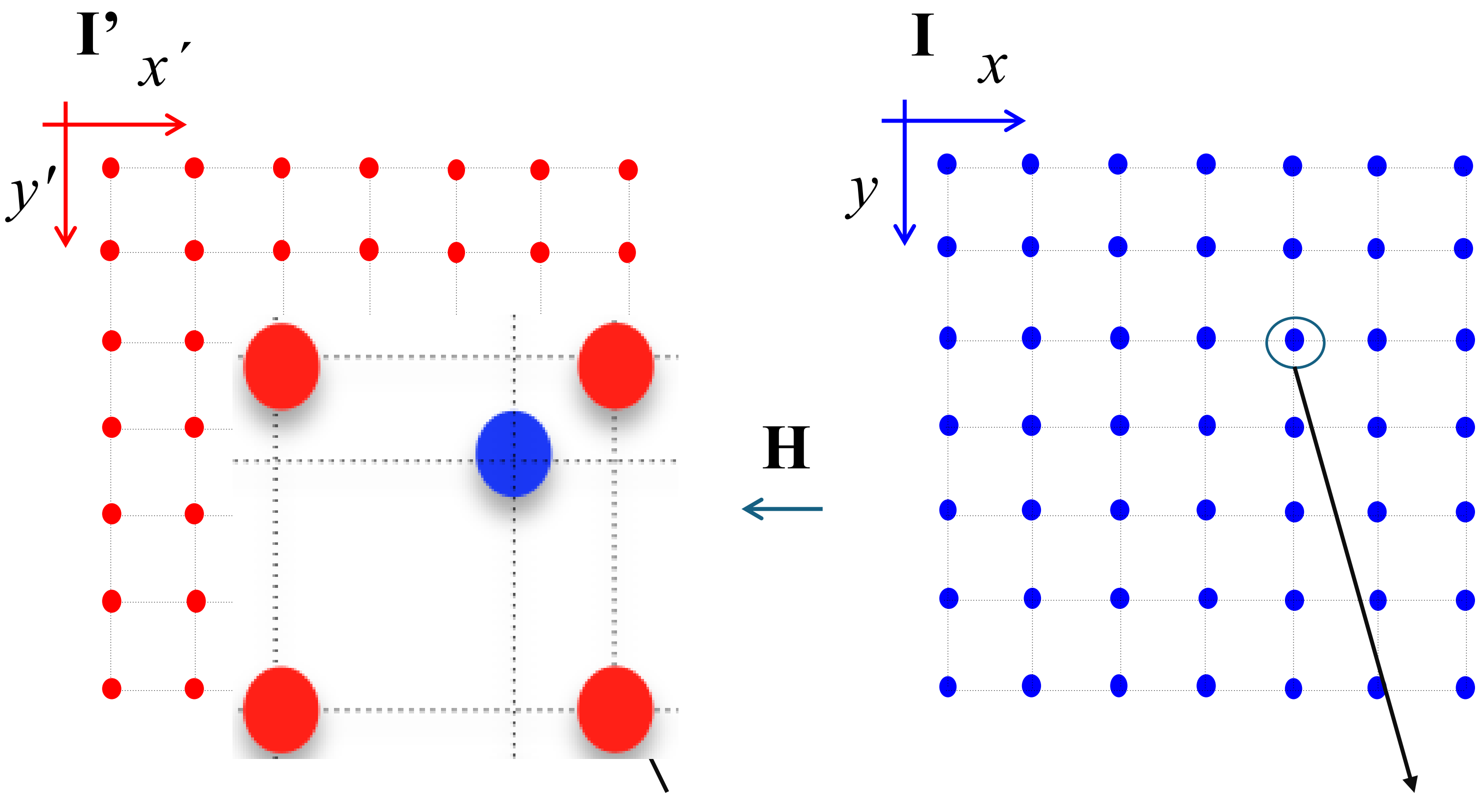
Coordinate Transformation: $$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

EXAMPLE



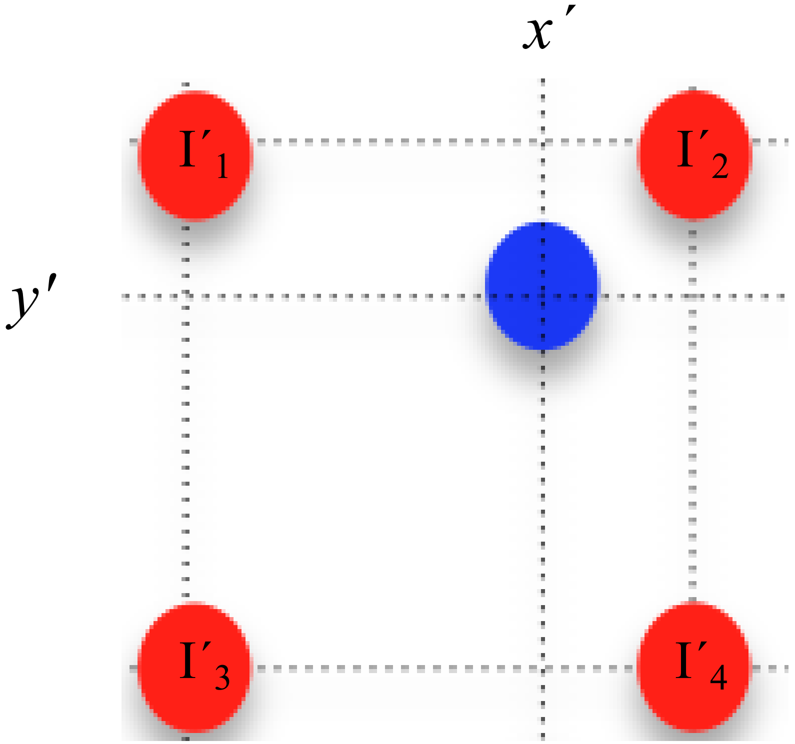
Coordinate Transformation: $$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

EXAMPLE



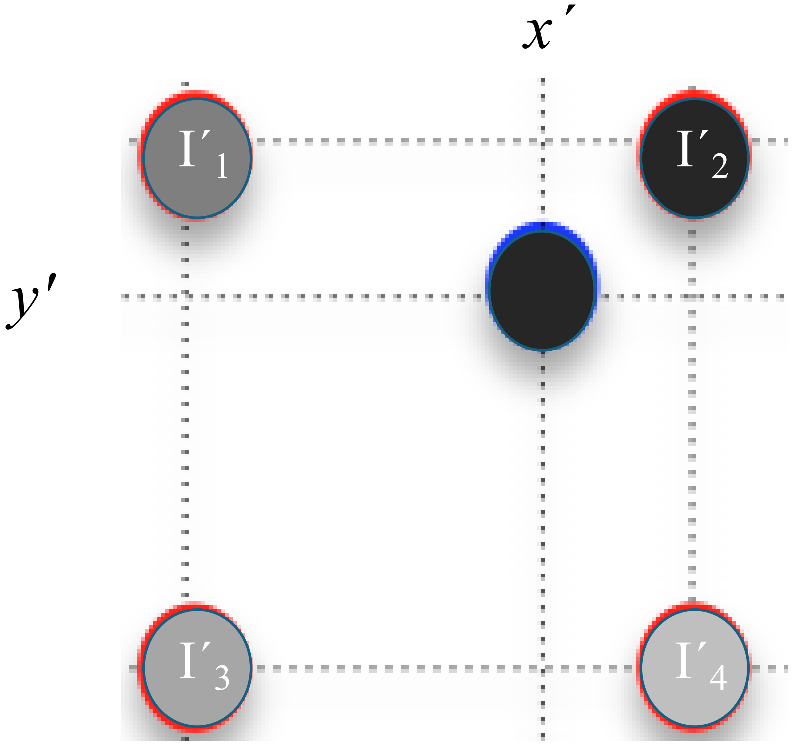
Coordinate Transformation:

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



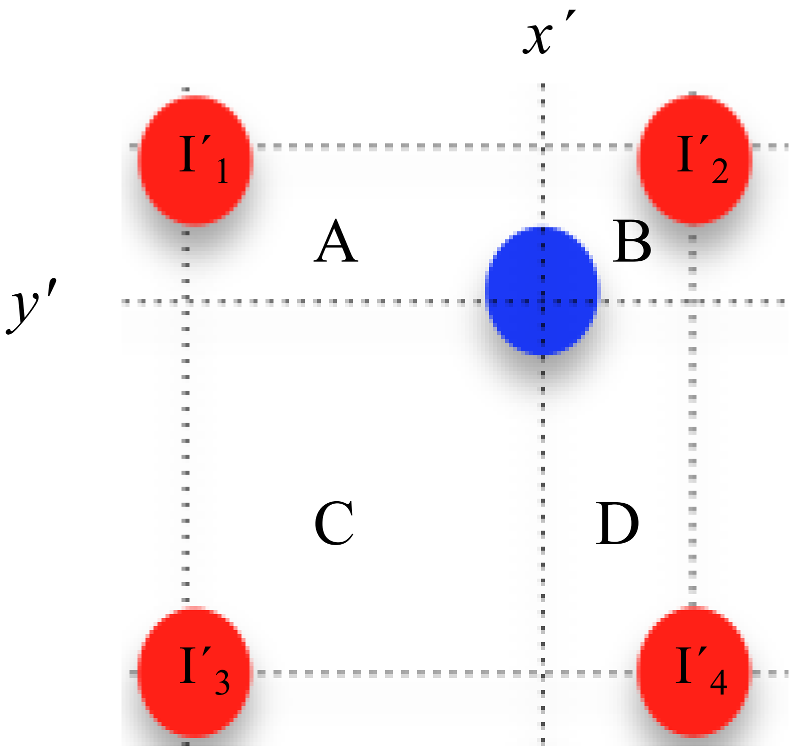
Nearest pixel

$\mathrm{I}(x,y) = \text{interpolation } \{\mathrm{I}'(x',y')\} = \mathrm{I}_2$



Nearest pixel

$I(x,y)$ = interpolation $\{I'(x',y')\} = I_2$

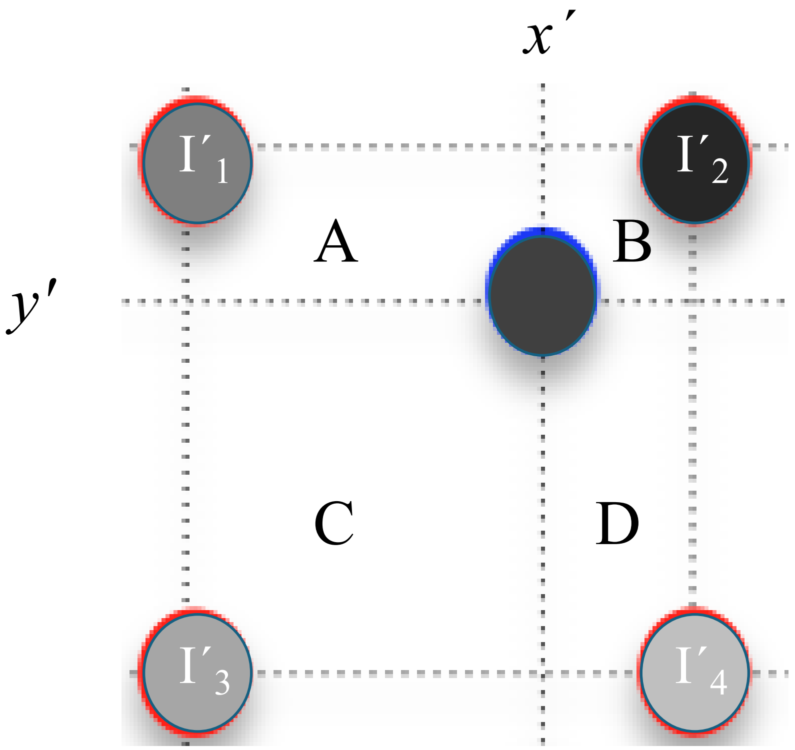


Bilinear interpolation

$I(x,y) = \text{interpolation } \{I'(x',y')\} =$

$$AI'_4+BI'_3+CI'_2+DI'_1$$

$A+B+C+D = 1$



Bilinear interpolation

$$I(x,y) = \text{interpolation } \{I'(x',y')\} =$$

$$AI'_4+BI'_3+CI'_2+DI'_1$$

$$A+B+C+D = 1$$

Implementation:

1. $x = 1, ..., N$; $y = 1, ..., M$:

2. $\mathbf{m} = [x \; y \; 1]^\mathsf{T}$.

3. $\mathbf{m}' = \mathbf{H}\mathbf{m}$.

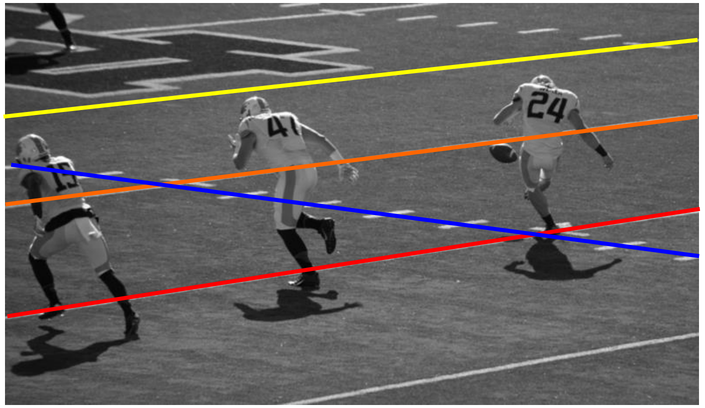
4. $x' = m_1/m_3$ y $y' = m_2/m_3$.

5. $\tilde{x} = \text{fix}(x')$, $\tilde{y} = \text{fix}(y')$,$\Delta x' = x' - \tilde{x}$ y $\Delta y' = y' - \tilde{y}$.

6. $$I(x, y) = [I'(\tilde{x}+1, \tilde{y}) - I'(\tilde{x}, \tilde{y})]\Delta x' + [I'(\tilde{x}, \tilde{y}+1) - I'(\tilde{x}, \tilde{y})]\Delta y' + [I'(\tilde{x}+1, \tilde{y}+1) + I'(\tilde{x}, \tilde{y}) - I'(\tilde{x}+1, \tilde{y}) - I'(\tilde{x}, \tilde{y}+1)]\Delta x' \Delta y' + I'(\tilde{x}, \tilde{y})$$

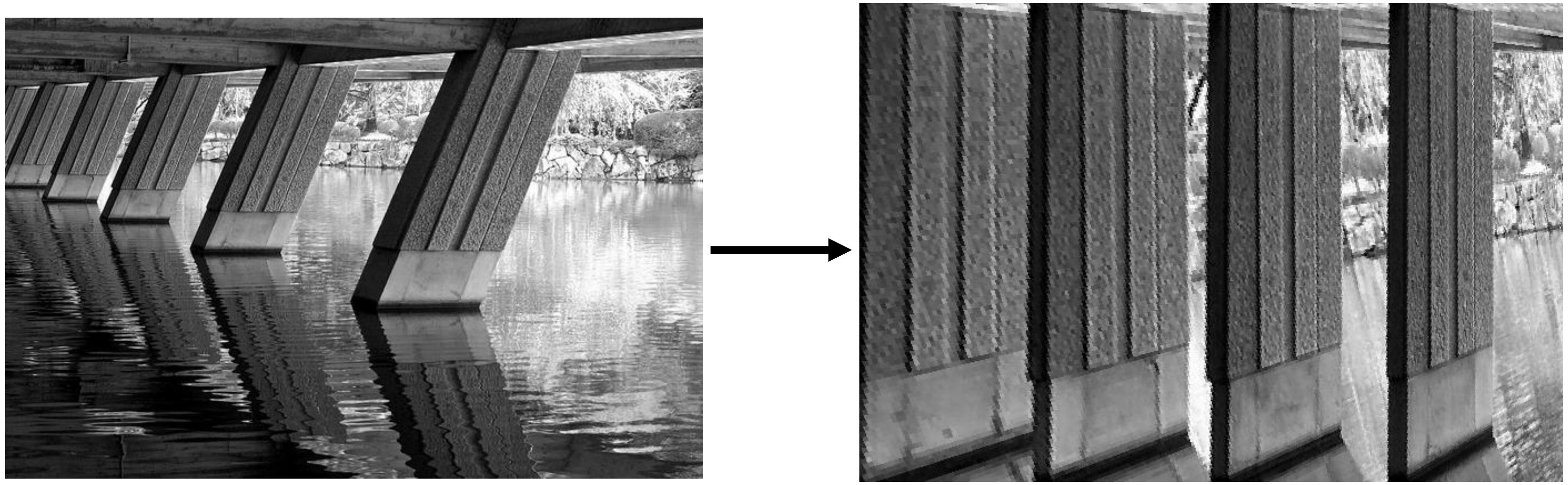
[Homography]

More examples.



[Homography]

More examples.



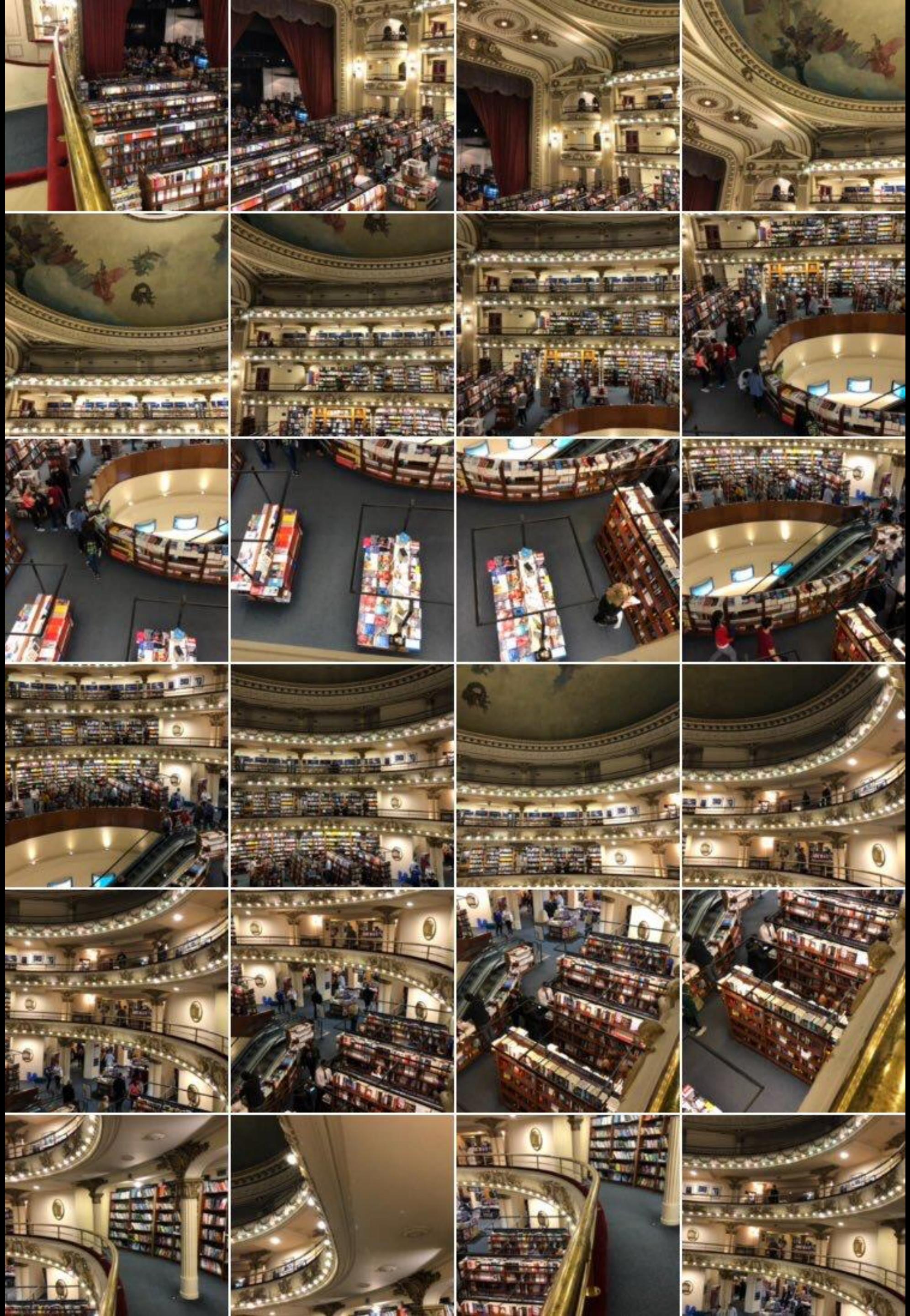
Mosaic Example 1

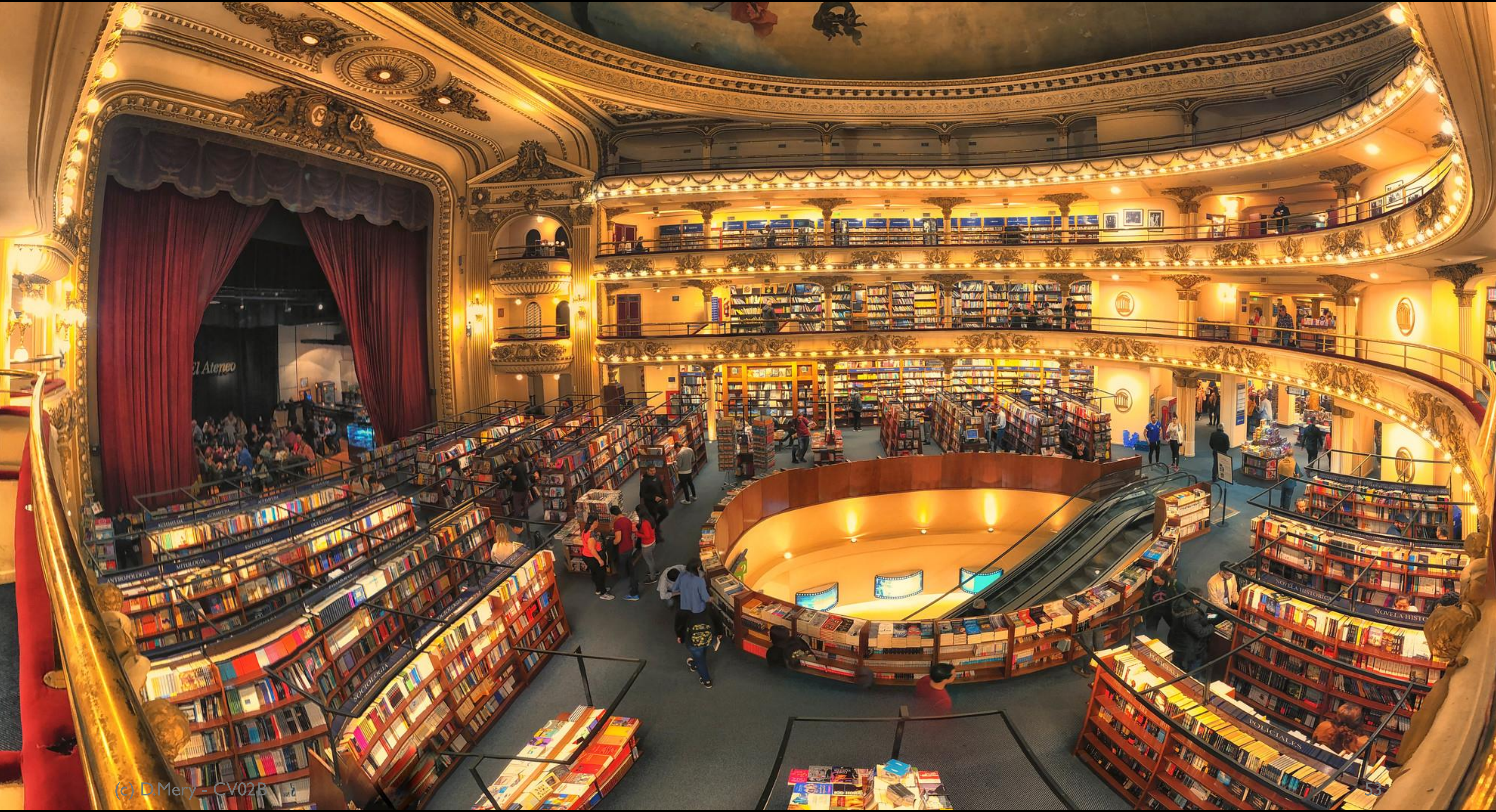
Rio de Janeiro

Mosaic Example 2

Buenos Aires

Librería Ateneo

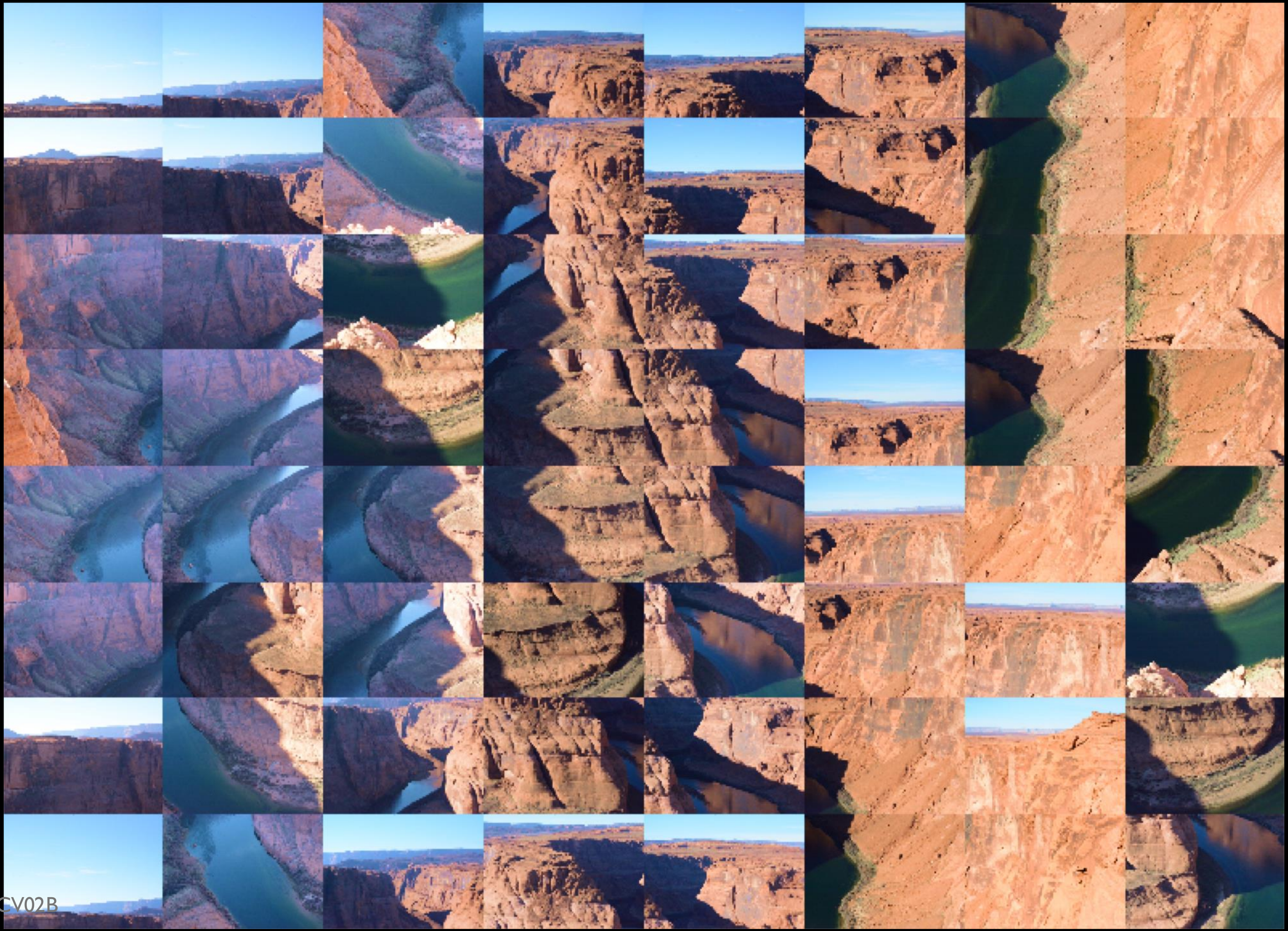



El Ateneo

Mosaic Example 3

Horseshoe Bend

Arizona



(c)